
ARMOR: Adaptive Retriever Optimization for Low-Resource Telecom Question Answering

Abstract

Telecom question answering (QA) is a challenging setting for retrieval-augmented generation (RAG): evidence is fragmented across standards, papers, encyclopedic resources, and web documents, and answers often hinge on technical tables, equations, and specialized protocol language. In low-resource subdomains, generator fine-tuning can over-specialize and degrade general capability, making query-side retriever adaptation an attractive alternative. To this end, we ask whether a fixed-generator, query-adapted RAG system can outperform generator-side adaptation, and which retriever objectives best support that setting. We argue that retrieval, not model fine-tuning, is an effective adaptation target: theoretically, we show that when domain data are scarce, query-encoder tuning can have lower estimation complexity than supervised fine-tuning under standard capacity assumptions. We identify two particularly relevant objectives—the latent-document RAG likelihood, which optimizes generation utility, and the InfoNCE contrastive objective, which improves semantic retrieval geometry—and leverage them jointly through a retriever optimization method targeting downstream QA performance in the telecom domain. Specifically, we introduce ARMOR, Adaptive Regularized Mixture Optimization for Retrievers, which learns separate temperatures for the RAG retrieval distribution and InfoNCE softmax and regularizes the adapted query encoder toward the frozen base query encoder. Across telecom-specific retrieval and generative QA benchmarks, we show that ARMOR improves evidence retrieval and answer generation in several in-domain settings.

1 Introduction

Telecommunications question answering (QA) is challenging for off-the-shelf LLMs because relevant evidence is fragmented across standards, research papers, encyclopedic resources, and web documents, while correct answers often depend on equations, tables, protocol logic, and cross-references. Recent resources such as Tele-Data and Tele-Eval make this setting tractable by pairing telecom corpora with QA supervision and provenance, naturally favoring grounded retrieval-augmented generation (RAG) over purely parametric knowledge storage [16]. In low-resource subdomains, however, full generator adaptation is brittle: labeled data are scarce, sequential tuning risks catastrophic forgetting [14], and prior telecom studies suggest that LoRA fine-tuning may provide limited improvements when the base model already only contains partial domain knowledge [8, 16]. On the other hand, RAG has emerged as a prominent paradigm for grounding LLM outputs in external knowledge sources [15], and has shown strong empirical performance in various applications.

We therefore study a simple yet important question: when data are scarce and the document index is fixed, can query-side retriever adaptation beat generator tuning, and which retrieval objective should drive it? Figure 1 previews the empirical answer, where QA performance for different methods is evaluated on Integrated Sensing and Communication (ISAC) domain. Our main claim is that low-resource telecom QA favors a retriever-centric strategy: keep the generator fixed, adapt the retriever, and leverage two complementary objectives. The latent-document RAG likelihood rewards evidence that improves answer generation [15], while InfoNCE improves semantic separation and

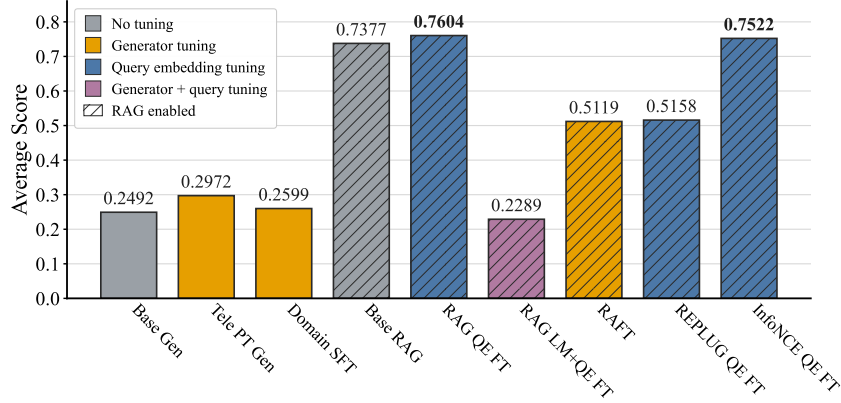


Figure 1: **Comparison of optimization targets and retriever-training objectives performance for ISAC domain QA** with Llama-3-8B-Instruct [6] as the generator model and e5-large-v2 [25] as the dense retriever backbone. Retriever-side query-encoder optimization produces substantially larger in-domain gains than either the base generator or generator-side adaptation, motivating our study of which component to optimize and which retriever objective best improves RAG performance. The compared methods are *Base Gen*, the closed-book generator without retrieval; *Tele PT Gen*, a telecommunications-domain pretrained generator model [16]; *Domain SFT*, a domain-specific supervised fine-tuned generator model; *Base RAG*, retrieval-augmented generation with the frozen base retriever; *RAG QE FT*, query-encoder fine-tuning under the original RAG marginal likelihood objective [15]; *REPLUG QE FT*, query-encoder fine-tuning using the REPLUG objective [21]; *RAFT*, query-encoder fine-tuning using the RAFT objective [26]; *InfoNCE QE FT*, query-encoder fine-tuning under the supervised contrastive InfoNCE objective [10].

retrieval discrimination [12, 10, 24]. We combine them through **ARMOR** (Adaptive Regularized Mixture Optimization for Retrievers), which introduces learnable temperatures into the RAG retrieval distribution and the InfoNCE softmax and adds query distillation to preserve compatibility with the frozen base document space.

[TC: You need to somehow highlight this central conclusion: For low-resource domain-specific RAG, the safest adaptation point is the query interface: ARMOR learns how sharply RAG likelihood and InfoNCE should shape the query encoder while regularizing it to remain compatible with a frozen document index.]

Contributions. We make the following contributions:

- C1) **We motivate empirically retriever-centric adaptation as a strong low-resource strategy for telecom QA.** We provide both theoretical and empirical evidence that, when domain supervision is scarce, query-encoder tuning can offer a better estimation–performance tradeoff than generator fine-tuning while preserving the modularity of a fixed RAG system.
- C2) **We introduce ARMOR, an adaptive objective-balancing method for retriever optimization.** ARMOR removes the need for a fixed mixture weight by learning separate softmax temperatures for RAG likelihood and InfoNCE, so each objective’s influence can change during training.
- C3) **We add base-compatible query regularization and validate the full method on telecom QA.** By distilling adapted query embeddings toward the frozen base query encoder, ARMOR limits drift from the fixed document embedding space. Across telecom retrieval and QA benchmarks, ARMOR gives the strongest overall tradeoff across in-domain retrieval quality, downstream answer quality, and robustness across generator scales.

These results suggest that low-resource domain adaptation for RAG systems should optimize the retrieval interface rather than the generator whenever the document index and generator remain stable.

2 What to Optimize in Low-Resource Telecom QA?

2.1 Query-Only RAG Setup and Retriever-Centric Adaptation

Let $\mathcal{D} = \{d_j\}_{j=1}^N$ be a datastore of chunked telecom passages derived from standards, research papers, and curated domain web sources. Given a telecom question x , the retriever returns a top- k set $\mathcal{N}_k(x) = \{d_i\}_{i=1}^k \subset \mathcal{D}$, and a generator produces an answer y conditioned on x and the retrieved evidence. We adopt a dense retrieval model with score

$$s_\eta(x, d) = f_\eta(x)^\top g(d),$$

where f_η is a trainable query encoder and g is a document encoder. In the most operationally attractive variant, g is fixed and the corpus is pre-embedded, so adaptation occurs only on the query side. We denote the retriever embedding dimension by d_E , so that $f_\eta(x), g(d) \in \mathbb{R}^{d_E}$.

This query-only setup is well matched to the low-resource telecom regime. In narrow telecom slices, the generator is often partially capable but fails because it is shown wrong or incomplete evidence. Updating the retriever is therefore a modular intervention that focuses limited supervision on the component controlling evidence exposure, while avoiding repeated re-indexing and preserving the base generator’s general capabilities.

This perspective is supported by the empirical pattern previewed in Figure 1: retriever-side optimization produces stronger in-domain gains than either the base model or generator-side fine-tuning, while better preserving broader capability. More importantly, retriever-centric learning aligns with operational constraints. It allows the system to update evidence access without repeatedly changing the generator, reuses a frozen document index when only the query encoder is tuned, and keeps grounding behavior interpretable through retrieved passages.

We next make this architectural choice more explicit by comparing the low-data generalization behavior of generator fine-tuning and retriever fine-tuning.

2.2 Low-Data Generalization Motivation

The next result is used only as a capacity comparison: under standard boundedness assumptions, the query-tuning class can have a smaller estimation term than the generator (*SFT*) when its effective dimension is smaller.

We compare two families of adaptation, *SFT* and *RAG*, on the same population distribution $\mathcal{P}$ over the data point $z = (x, y)$, where x is the question, and y is the target answer. Given N i.i.d. training triples $\{z_i\}_{i=1}^N$ drawn from the population distribution $\mathcal{P}$, let $\theta \in \mathcal{E}_S \subset \mathbb{R}^{d_S}$ be the trainable generator parameter for *SFT*, $\eta \in \mathcal{E}_R \subset \mathbb{R}^{d_R}$ be the trainable *RAG* parameters, and $\ell_S(z; \theta)$ and $\ell_R(z; \eta)$ be the per-sample *SFT* and *RAG* loss, respectively.

The population *SFT* loss is defined as the expected test loss under the true population distribution $\mathcal{P}$, while the empirical *SFT* loss is defined as the empirical training loss on finite sample $\{z_i\}_{i=1}^N$, i.e.

$$\text{population loss: } L_S(\theta) = \mathbb{E}_{z \sim \mathcal{P}} [\ell_S(z; \theta)], \quad \text{empirical loss: } \hat{L}_{S,N}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell_S(z_i; \theta).$$

Similarly, we can define the population and empirical *RAG* losses as follows.

$$\text{population loss: } L_R(\eta) = \mathbb{E}_{z \sim \mathcal{P}} [\ell_R(z; \eta)], \quad \text{empirical loss: } \hat{L}_{R,N}(\eta) = \frac{1}{N} \sum_{i=1}^N \ell_R(z_i; \eta).$$

The goal of generalization theory is to bound the gap between the population loss and the empirical loss, where a smaller bound indicates better generalization ability of the empirical optimal solution.

We denote the probability space for *RAG* answer as $q_\eta(\cdot | x) = \sum_{d \in \mathcal{N}_k(x)} p_\eta(d | x) p_{\theta_0}(\cdot | x, d)$. Therefore, letting $\ell_{\text{NLL}}(q, x, y) = -\log q(y | x)$ be the negative log-likelihood loss, the *SFT* and *RAG* loss can be rewritten as

$$\ell_S(z; \theta) = \ell_{\text{NLL}}(p_\theta, x, y), \quad \text{and} \quad \ell_R(z; \eta) = \ell_{\text{NLL}}(q_\eta, x, y).$$

where the first optimizes the generator parameter θ to predict the answer y given the question x , and the latter optimizes the retrieval parameter η to select the relevant document d , so that the model can generate the correct answer y conditioned on the resulting question–document pair (x, d) .

[TC: The theory section may invite technical scrutiny if readers expect Theorem 1 to explain the exact empirical gains rather than serve as a capacity-comparison motivation. May make the theorem low-profile.]

Theorem 1 (Generalization theory for SFT and RAG). *Assume that the loss function $\ell_{\text{NLL}}(q, x, y)$ is bounded in $[0, c]$ and is ρ -Lipschitz w.r.t q in the feasible domain, and the weights for SFT and RAG are bounded by R , for all t . Also assume that the tokenized inputs for question and answers (x_n, y_n) have full rank and is bounded by R_X scaled with the corresponding dimension d_R and d_S , for all $n \in [N]$. Additionally, assume $\|g(d)\| \leq B_g$ and q_η is ρ^{soft} -Lipschitz over $s_\eta(x, d)$ on the feasible domain. Then with probability at least $1 - \delta$, we have for any $\theta \in \mathcal{E}_S$ and $\eta \in \mathcal{E}_R$,*

$$L_S(\theta) \leq \hat{L}_{S,N}(\theta) + \tilde{O}(\rho C(N, R, R_X, d_S, d_m)) + 3c\sqrt{\frac{\log(2/\delta)}{2N}} \quad (1)$$

$$L_R(\eta) \leq \hat{L}_{R,N}(\eta) + \tilde{O}(\rho^{\text{soft}} B_g \sqrt{d_E} \rho C(N, R, R_X, d_R, d_{m'})) + 3c\sqrt{\frac{\log(2/\delta)}{2N}} \quad (2)$$

where $C(N, R, R_X, d, d_m) := \sqrt{\frac{P(d, d_m)}{N^3}} \left(1 + \log \left(R_O R_V (\sqrt{d} R_X) \sqrt{\frac{N}{P(d, d_m)}}\right)\right)$ and

$$P(d, d_m) = (\sqrt{d} R_X)^2 \left(\left(\sqrt{d_m} R_V\right)^{\frac{2}{3}} + \left(\sqrt{d_m} R_K R_Q R_V\right)^{\frac{2}{3}} \right)^3 \log(Nd)$$

with d_m and $d_{m'}$ denoting the corresponding model architecture widths for SFT and RAG.

Theorem 1 is built upon the generalization theory for Transformer based model [19, 23] and the retrieval tuning [3]. The proof of Theorem 1 is provided in Appendix B.

Remark 1. *In the sample-limited regime, N is small, so the second complexity term dominates both generalization error for SFT and RAG, compared with the difference between the empirical risks. Omitting the logarithmic factors, the SFT complexity term scales as $\tilde{O}(\rho \frac{\sqrt{d_S d_m}}{N^{3/2}})$, whereas the RAG complexity term scales as $\tilde{O}(\rho \rho^{\text{soft}} B_g \frac{\sqrt{d_E d_R d_{m'}}}{N^{3/2}})$. Therefore, when $\rho^{\text{soft}} B_g = O(1)$, the RAG bound is tighter whenever $d_E d_R d_{m'} \lesssim d_S d_m$.*

In our experiments, query-encoder adaptation uses about four times fewer trainable parameters than generator adaptation, which suggests $d_E d_R d_{m'} \lesssim d_S d_m$. Therefore, Theorem 1 provides theoretical support for the empirical evidence in Figure 1 on why retriever tuning can be preferable to generator tuning when data are scarce. The remaining design question is which retriever objectives should drive this adaptation.

2.3 Retriever Objectives: RAG Likelihood, InfoNCE, and Static Mixing

Following the empirical pattern in Figure 1, we focus on two complementary retriever objectives: RAG likelihood and InfoNCE.

RAG likelihood. With a frozen generator $p_{\theta_0}(y \mid x, d)$, the RAG objective optimizes retrieval indirectly through answer likelihood:

$$\mathcal{L}_{\text{RAG}}(x, y) = -\log \sum_{d \in \mathcal{N}_k(x)} p_\eta(d \mid x) p_{\theta_0}(y \mid x, d),$$

where $\mathcal{N}_k(x)$ is the top- k document set ranked by the retrieval score $s_\eta(x, d)$ and

$$p_\eta(d | x) = \frac{\exp(s_\eta(x, d))}{\sum_{d' \in \mathcal{N}_k(x)} \exp(s_\eta(x, d'))}.$$

This objective rewards passages that are useful for generation, not merely passages that are topically related.

InfoNCE. Given a positive passage d^+ and a set of negatives $\mathcal{N}^-$, the contrastive objective is

$$\mathcal{L}_{\text{InfoNCE}}(x, d^+, \mathcal{N}^-) = -\log \frac{\exp(s_\eta(x, d^+))}{\exp(s_\eta(x, d^+)) + \sum_{d^- \in \mathcal{N}^-} \exp(s_\eta(x, d^-))}.$$

In telecom QA, positive passages can often be obtained from source-linked datasets such as Tele-Eval, while negatives may come from in-batch sampling, BM25, or hard-negative mining. This objective primarily improves semantic separability and retrieval discrimination.

These objectives matter for different reasons. RAG likelihood is utility-driven: it improves the retriever to the extent that retrieval helps the generator answer correctly. InfoNCE is geometry-driven: it improves the embedding space so that relevant evidence is easier to recover. In low-resource telecom QA, both signals are needed. RAG alone may under-shape the representation space, while InfoNCE alone may reward topical similarity without fully aligning retrieval with downstream answer utility. Empirically, both objectives perform well in Figure 1, which motivates combining them rather than treating either objective as sufficient on its own.

A standard mixture writes

$$\mathcal{L}_{\text{static}} = \lambda \mathcal{L}_{\text{RAG}} + (1 - \lambda) \mathcal{L}_{\text{InfoNCE}},$$

where $\lambda \in [0, 1]$ is fixed in advance. The static mixture has two limitations that ARMOR targets directly: it fixes the objective balance for all training stages, and it does not constrain the adapted query encoder to remain compatible with the frozen document index.

3 ARMOR: Adaptive and Regularized Retriever Optimization

3.1 Temperature-Modulated Retriever Objectives

The fixed mixture in $\mathcal{L}_{\text{static}}$ assumes that the relative influence of RAG and InfoNCE should remain constant throughout training. This is a strong assumption: the two objectives capture different aspects of retriever learning, and their relative usefulness need not be the same early and late in optimization. Kendall et al. [13] make a related point in multi-task learning, showing that fixed loss weighting can bias training toward one objective and that adaptive weighting can lead to more balanced optimization.

Our setting differs in that we optimize a single retriever with two complementary signals, and we place adaptivity inside the softmax distributions that define the objectives rather than on external loss coefficients. We therefore introduce learnable temperatures into both retriever objectives. For RAG, the retriever distribution over the top- k documents becomes

$$p_\eta^{(\tau_r)}(d_i | x) = \frac{\exp(s_\eta(x, d_i)/\tau_r)}{\sum_{j=1}^k \exp(s_\eta(x, d_j)/\tau_r)},$$

which yields the temperature-modulated loss

$$\mathcal{L}_{\text{RAG}}(x, y; \tau_r) = -\log \sum_{i=1}^k p_\eta^{(\tau_r)}(d_i | x) p_{\theta_0}(y | x, d_i).$$

For InfoNCE, we similarly define

$$\mathcal{L}_{\text{InfoNCE}}(x, d^+, \mathcal{N}^-; \tau_c) = -\log \frac{\exp(s_\eta(x, d^+)/\tau_c)}{\exp(s_\eta(x, d^+)/\tau_c) + \sum_{d^- \in \mathcal{N}^-} \exp(s_\eta(x, d^-)/\tau_c)}.$$

Large temperatures produce smoother distributions and broader supervision; small temperatures sharpen the objectives and focus training on top-ranked or hardest competing items. We then define the adaptive retriever objective as

$$\mathcal{L}_{\text{mix}} = \mathcal{L}_{\text{RAG}}(x, y; \tau_r) + \mathcal{L}_{\text{InfoNCE}}(x, d^+, \mathcal{N}^-; \tau_c).$$

3.2 Gradient Interpretation of Adaptive Temperature

Adaptive temperatures can be viewed as objective-side parameters that control how strongly the RAG and InfoNCE losses shape the query encoder. This explains how temperature learning acts as an implicit adaptive weighting mechanism, even though the final ARMOR objective does not attach explicit scalar mixture weights to the two losses.

We first consider the contrastive objective in a minimal two-document setting with one positive document d^+ , one negative document d^- , and query embedding q . Let $s^+ = q^\top d^+$ and $s^- = q^\top d^-$, and define the margin $\Delta = s^+ - s^-$. The InfoNCE loss with temperature τ_c is

$$\mathcal{L}_{\text{NCE}}(q, \tau_c) = \log(1 + \exp(-\Delta/\tau_c)).$$

Differentiating with respect to the query embedding yields

$$\nabla_q \mathcal{L}_{\text{NCE}} = -\frac{1}{\tau_c} \sigma\left(-\frac{\Delta}{\tau_c}\right) (d^+ - d^-),$$

where $\sigma(\cdot)$ is the logistic sigmoid. Thus, τ_c directly scales the gradient through $1/\tau_c$ and controls how sharply updates focus on hard-margin examples. Smaller τ_c makes the contrastive loss act more aggressively on the query encoder, while larger τ_c smooths its influence.

An analogous effect appears in the retrieval component of the RAG objective. Consider a top-2 setting with score gap $\Delta = s_1 - s_2$ between two retrieved documents. The temperature-modulated retrieval distribution is $p_1 = \sigma(\Delta/\tau_r)$ and $p_2 = 1 - p_1$, giving the RAG loss

$$\mathcal{L}_{\text{RAG}}(q, \tau_r) = -\log(p_1 a_1 + p_2 a_2),$$

where a_1 and a_2 denote the generator-side answer utilities of the two retrieved documents. Differentiating with respect to the score gap again yields an explicit $1/\tau_r$ factor. Hence, smaller retrieval temperature makes the loss more sensitive to differences among top-ranked documents, while larger temperature spreads the training signal more broadly.

Together, these observations show that temperatures do more than change softmax entropy: they determine how strongly each objective contributes gradient signal to the query encoder. In this sense, τ_r and τ_c act as learned, objective-specific weighting mechanisms. Decreasing a temperature sharpens the corresponding objective and increases its local influence, while keeping it large makes the objective more conservative. This view explains how adaptive temperatures can replace manually tuned mixture weights while still allowing the relative influence of RAG likelihood and InfoNCE to evolve during training.

The main consequence of this formulation is that adaptive temperature learning provides a mechanism for *balanced retriever optimization*. Rather than fixing the sharpness of the two objectives in advance, the model can begin training in a smoother regime and progressively sharpen one or both objectives as the retriever becomes more reliable. This makes the approach particularly appealing in low-resource settings, where fixed weighting can otherwise be brittle and highly sensitive to tuning.

At the same time, the same mechanism also introduces a potential failure mode. Once the retriever is already locally correct on many training examples, gradient descent can continue driving the temperatures downward, eventually making the softmaxes excessively sharp. In that regime, temperature learning no longer improves balance, but instead acts as a shortcut that amplifies existing score differences. This observation motivates the regularization strategy introduced below, which constrains the query encoder to remain compatible with the frozen base document space even as the retriever objectives become sharper.

3.3 Query Distillation and Final ARMOR Objective

Only the query encoder is updated in our setting, while the document encoder and index remain fixed in the base embedding space. As a result, improving the mixed retriever objective alone can still move the query encoder away from the geometry used at inference time. This issue is especially acute in low-resource domains, where optimization can over-specialize to the training objective.

To preserve compatibility with the frozen document space, we regularize the adapted query encoder toward the frozen base query encoder. For a minibatch of queries, we use cosine query distillation,

$$\mathcal{L}_{\text{qdist}} = \frac{1}{B} \sum_{b=1}^B (1 - \cos(q_\eta(x_b), q_0(x_b))),$$

where $q_\eta(x_b)$ is the adapted query embedding and $q_0(x_b)$ is the corresponding frozen base embedding. Because both query and document embeddings are ℓ_2 -normalized, cosine similarity provides a natural measure of deviation from the original retrieval geometry.

The adaptive temperatures and the query-distillation regularizer together define our full training framework, **ARMOR** (Adaptive Regularized Mixture Optimization for Retrievers). ARMOR combines adaptive mixture optimization of the RAG and InfoNCE objectives with regularization that preserves compatibility with the frozen base retriever space. The resulting objective is

$$\mathcal{L}_{\text{ARMOR}} = \mathcal{L}_{\text{RAG}}(x, y; \tau_r) + \mathcal{L}_{\text{InfoNCE}}(x, d^+, \mathcal{N}^-; \tau_c) + \mathcal{L}_{\text{qdist}}$$

where the adaptive temperatures determine how selectively the utility-driven and geometry-driven objectives shape the query encoder, and the query distillation prevents this flexibility from degenerating into excessive drift away from the pretrained retrieval space.

4 Related Work

Retrieval-Augmented Architectures. Retrieval-augmented systems use external evidence as non-parametric memory to complement model weights. REALM and RAG connected retrieval to downstream learning by treating documents as latent variables [7, 15], while FiD, Atlas, RETRO, and in-context retrieval-augmented language models established the benefits of modular memory, multi-passage reasoning, and updatable indices [9, 11, 4, 2]. REPLUG is closest to our framing because it keeps the language model fixed and adapts prediction through retrieval [21]. However, as discussed in Section 1, REPLUG-style prediction can incur additional test-time overhead from document-wise scoring or aggregation, whereas our method retains the standard RAG-style retrieval-and-generation interface of Lewis et al. [15] while modifying the retriever-training objective to improve downstream performance.

Dense Retrieval and Contrastive Learning. Dense passage retrieval introduced the dual-encoder paradigm for open-domain QA, embedding queries and documents in a shared space with contrastive supervision [12]. Contriever shows that such objectives can produce strong retrievers with limited labeled data, making them well suited to low-resource adaptation [10]. InfoNCE formalizes this discrimination-based training by contrasting positives and negatives, with temperature controlling softmax sharpness [24]. In contrast to purely contrastive retriever adaptation, we combine contrastive learning with generation-focused retrieval objectives and use adaptive temperature control to balance their influence during retriever fine-tuning.

Telecom Grounding and Evaluation. Tele-Data and Tele-Eval provide telecom-specific corpora, QA supervision, and source identifiers for grounded QA [16]. KILT highlights the need to evaluate both answer correctness and evidence recovery in knowledge-intensive tasks [20], while RAGAS offers reference-free measures of faithfulness and context relevance when exact citations are incomplete or noisy [5]. This is critical in telecom, where plausible but unsupported answers are insufficient. Building on these resources, we identify which components of a telecom RAG system should be adapted for domain-specific knowledge while limiting overfitting in low-resource settings.

Regularization and Objective Balancing. Multi-task learning balances heterogeneous objectives with static or uncertainty-based weights [13], and continual-learning methods address catastrophic forgetting during adaptation [14]. In contrast, our setting updates only the query encoder while keeping the document encoder and index frozen, requiring both effective combination of RAG likelihood and InfoNCE and compatibility with the frozen retrieval space. This motivates regularization that limits parameter drift, preserves base query embeddings, or maintains base ranking behavior.

5 Experiments

We evaluate ARMOR on domain-specific telecommunications QA tasks along three axes: gains over retriever-training baselines, training dynamics of the adaptive and regularized objective, and robustness across generator scale and model family.

[TC: 1. Add an ablation table that separates (i) RAG only, (ii) InfoNCE only, (iii) static mix, (iv) adaptive temperatures without query distillation, (v) query distillation without adaptive temperatures, and (vi) full ARMOR. Report the mean and standard deviation across seeds.

Method	Tele-eval											
	ISAC				JCC				SAGIN			
	Score	R@1	R@3	R@5	Score	R@1	R@3	R@5	Score	R@1	R@3	R@5
Base Gen	0.2492	—	—	—	0.1895	—	—	—	0.1808	—	—	—
Base RAG	0.7377	0.9300	0.9750	0.9850	0.6643	0.8250	0.9350	0.9650	0.6314	0.8300	0.9500	0.9750
RAG QE FT	0.7604	0.9350	0.9850	0.9900	0.6512	0.8500	0.9600	0.9750	0.6947	0.8700	0.9550	0.9800
InfoNCE QE FT	0.7522	0.9450	0.9650	0.9800	0.6572	0.8300	0.9300	0.9450	0.6341	0.8550	0.9500	0.9750
Mix QE FT	0.7626	0.9400	0.9800	0.9800	0.6654	0.8550	0.9450	0.9700	0.6786	0.8400	0.9600	0.9750
ARMOR	0.7637	0.9450	0.9750	0.9800	0.6683	0.8400	0.9550	0.9750	0.6520	0.8650	0.9550	0.9750

Table 1: Tele-Eval open-ended QA and retrieval results across ISAC, JCC, and SAGIN. Score is the LLM-judge average answer score; R@ k is source-document recall at rank k . Best values within each domain and metric are bolded.

2. Add a train-size experiment with 5 to 6 points, for example $n \in \{10, 25, 50, 100, 200, \dots\}$ per domain when feasible. Plot Base RAG, RAG QE FT, Mix QE FT, and ARMOR on Tele-Eval score and at least one retrieval metric.
3. Add a small qualitative failure/success analysis, especially for ISAC TeleQnA transfer. Include 3 to 5 qualitative examples for illustrative purposes. Turns a weak transfer result into insight.
4. Split the experimental claims into two rows: training-aligned retrieval/generation and out-of-corpus telecom transfer. Explain that ARMOR is designed to reduce over-specialization, then show where it succeeds and where it does not.]

5.1 Experimental Setup

We evaluate ARMOR on domain-specific telecommunications QA tasks, focusing primarily on three representative 6G subdomains: Integrated Sensing and Communication (ISAC), Joint Communication and Computation (JCC), and Space-Air-Ground Integrated Network (SAGIN). For each domain, we construct train/validation/test splits from a curated telecommunications corpus using a multi-stage filtering, chunking, and grounded QA generation pipeline. ARMOR is compared against several retriever-training baselines. For clarity, we use the following shorthand in tables and figures. **Base Gen** denotes the closed-book generator without retrieval, and **Base RAG** denotes retrieval-augmented generation with the frozen base retriever. **RAG QE FT** denotes query-encoder fine-tuning under the RAG objective, **InfoNCE QE FT** denotes query-encoder fine-tuning under the supervised contrastive InfoNCE objective, and **Mix QE FT** denotes non-adaptive mixed-objective query-encoder fine-tuning. Our proposed method is denoted **ARMOR**. Across all experiments, the dense retriever backbone is `intfloat/e5-large-v2`; document embeddings are kept fixed in the base embedding space, while only the query encoder is adapted. For answer generation, we use **Llama-3-8B-Instruct** as the primary generator in most experiments, and additionally study **Llama-3.2-1B**, **Llama-3.2-3B**[6], and **Qwen3-8B** [22] to analyze how the effect of retriever optimization varies across generator scale and model family. Evaluation is conducted using two complementary benchmarks: **Tele-Eval** [16], which measures open-ended QA quality and retrieval fidelity through an LLM-as-a-Judge average score and Recall@{1, 3, 5}, and **TeleQnA** [17], which measures multiple-choice QA accuracy. Tele-Eval is partially aligned with the corpus used for retriever adaptation, whereas TeleQnA provides a stronger test of out-of-corpus generalization. Full details of data construction, model configurations, baselines, and evaluation are provided in Appendix A.1, Appendix A.2, Appendix A.3, and Appendix A.4.

5.2 Experimental Results

Unless otherwise noted, we use the compact baseline names introduced in Section 5.1: **Base Gen**, **Base RAG**, **RAG QE FT**, **InfoNCE QE FT**, **Mix QE FT**, and **ARMOR**.

Domain-Specific RAG Performance. Tables 1 and 2 show that the benefits of retriever fine-tuning depend on whether evaluation is aligned with the training corpus. On Tele-Eval, which is built from the same underlying corpus as retriever training, Tele-Data dataset [16], ARMOR is the most consistent method overall, giving the best score on ISAC and JCC, while RAG QE FT is strongest on SAGIN; all query-encoder fine-tuning methods improve over Base RAG on at least some retrieval metrics. On TeleQnA, which is not drawn from the training corpus, the picture is more mixed: Base

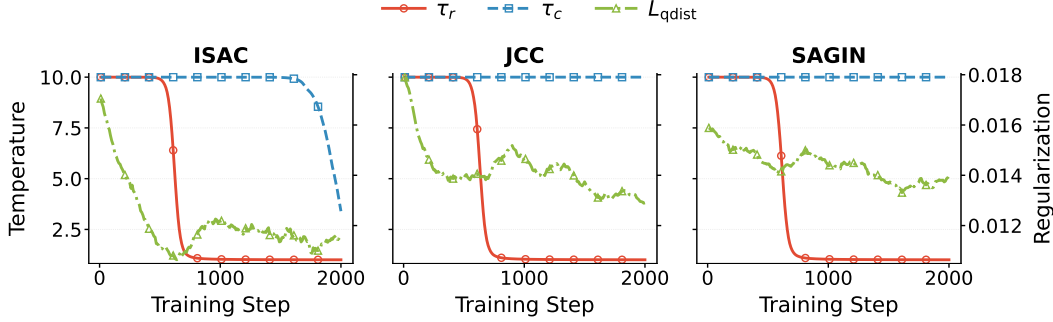


Figure 2: Training dynamics of adaptive temperatures and query-distillation regularization across ISAC, JCC, and SAGIN. In all three domains, the retrieval temperature stays high early and then drops sharply, while the query-distillation loss rises again late in training. Only ISAC shows a visible decrease in the InfoNCE temperature, suggesting that retrieval-side sharpening is robust across domains whereas contrastive sharpening is more domain-dependent.

RAG remains best on ISAC, ARMOR ties the best result on JCC, and ARMOR is best on SAGIN. The weaker ISAC generalization is plausibly due to its much smaller training set, which makes in-corpus adaptation easier to overfit. Overall, ARMOR provides the most robust tradeoff across domains, especially in the large dataset.

ARMOR Training Dynamics. Figure 2 shows a consistent interaction between adaptive temperatures and query regularization. In all three domains, the retrieval temperature stays high early, then drops sharply toward its minimum, while the query-distillation loss rises again late in training, indicating increasing tension with the frozen base embedding space. The two temperatures evolve on different timescales: retrieval sharpens earlier and more strongly than InfoNCE, and only ISAC shows a visible late-stage decrease in the InfoNCE temperature. Thus, retrieval-side sharpening is a robust cross-domain pattern, while contrastive sharpening appears more domain-dependent.

Method	Tele-QnA		
	ISAC	JCC	SAGIN
Base Gen	0.7000	0.600	0.6600
Base RAG	0.7400	0.6400	0.6800
RAG QE FT	0.7000	0.6600	0.7000
InfoNCE QE FT	0.7000	0.6400	0.7000
Mix QE FT	0.7000	0.6200	0.6400
ARMOR	0.6800	0.6600	0.7600

Table 2: Tele-QnA accuracy across domains. Best results are highlighted in bold.

Effect of Generator Scale. Figure 3 shows that the gain of ARMOR over Base RAG diminishes as the generator becomes stronger. The improvement is large for Llama-3.2-1B, but much smaller for Llama-3.2-3B, suggesting that stronger generators already capture more of the benefit of retrieval. Qwen3-8B slightly outperforms the similarly sized Llama baseline under both Base RAG and ARMOR, indicating that model family matters in addition to parameter count. Overall, retriever optimization is most valuable in the low-capacity regime, but remains beneficial for stronger generators.

6 Conclusion

We studied low-resource telecom QA and showed that retriever-side adaptation can be a more effective and operationally stable alternative to generator fine-tuning. We proposed **ARMOR**, which combines RAG likelihood and InfoNCE through adaptive temperature-based objective shaping, while using query distillation to preserve compatibility with the frozen document embedding space. Empirically, ARMOR provides a robust tradeoff between retrieval specialization and downstream QA performance, with training dynamics showing that adaptive temperatures sharpen the retrieval signal while regularization limits query-encoder drift. These results suggest that low-resource RAG adaptation should jointly optimize complementary retriever objectives under compatibility constraints.

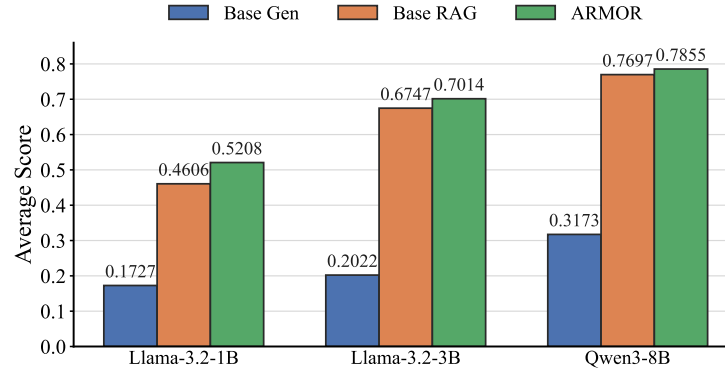


Figure 3: Comparison of Base Gen, Base RAG, and ARMOR across generator backbones. ARMOR yields the largest gain for the smallest model and smaller gains for stronger generators, indicating that retriever optimization is most valuable when generator capacity is limited. Qwen3-8B slightly outperforms the similarly sized Llama baseline under both Base RAG and ARMOR, showing that model family matters in addition to scale.

The main limitation is that gains are strongest when evaluation is aligned with the adaptation corpus; out-of-corpus transfer is mixed and should be validated with repeated runs and uncertainty estimates. Because telecom QA can support technical decision-making, future work should also quantify citation reliability and failure modes before deployment.

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A Experiment Details

In this section, we provide additional details on the data generation, baseline implementation, and experiment setup used in this paper. The codebase used for experiments in this paper can be found here: https://anonymous.4open.science/r/ARMOR_NeurIPS2026-C1DB/. All experiments were conducted on a server equipped with 8 NVIDIA A100 80GB SXM GPUs.

A.1 Data Generation and Pre-processing Pipeline

The data generation process transforms raw telecommunications data into a high-quality, domain-specific dataset. The pipeline is used to construct training and evaluation sets for five telecommunications domains: Integrated Sensing and Communication (ISAC), Joint Communication and Computation (JCC), Reconfigurable Intelligent Surfaces (RIS), Space-Air-Ground Integrated Networks (SAGIN), and Semantic Communication (SemCom).

Domain definitions. We focus on five key 6G research pillars. ISAC refers to the co-design of wireless communication and radar sensing systems that share hardware, spectrum, and waveforms, such as OFDM-based sensing. JCC, also referred to as Joint Communication and Computing, focuses on the tight coupling of radio resource allocation and edge/cloud computation, such as mobile-edge-computing task offloading. RIS refers to programmable metasurfaces that control electromagnetic wave propagation to create smart radio environments, mitigate blockages, and enhance coverage. SAGIN refers to the vertical integration of terrestrial networks with non-terrestrial segments, including LEO/MEO/GEO satellites, high-altitude platform stations, and unmanned aerial vehicles. SemCom refers to semantic communication, a paradigm that shifts the objective from bit-accurate transmission to meaning-accurate communication, optimizing for task success or intent recovery rather than raw data reconstruction.

Corpus statistics. The source corpus is the Tele-Data dataset [16], which contains approximately 2.5B tokens, or 12.1GB, aggregated from arXiv, 3GPP standards, Wikipedia, and telecommunications-related web crawls. Table 3 summarizes the filtering and generation statistics for a representative subset of domains. The high example-to-document ratio is due to the dense technical nature of the retained documents, such as 3GPP technical specifications, which often span hundreds of pages.

Multi-step document filtering. Documents are filtered using a multi-step process to ensure both high recall and high precision. First, a lightweight keyword prefilter scans the first 6,000 characters and available metadata, including title, URL, and standards series, for domain-specific keywords. For example, the ISAC filter targets terms such as *ISAC*, *JCAS*, *radar-communication*, and 3GPP series 36/37/38.

Second, documents passing the prefilter are scored by a local instruction-tuned large language model using a Boolean relevance flag and a confidence score in $[0, 1]$. A document is retained if it is marked relevant and has confidence at least 0.70.

Third, an optional high-precision post-filter is applied using a stronger external language model through an API. This stage uses a relevance threshold of 0.50 to remove fringe or non-technical content, such as market news, that may have passed the local filter. The refinement model is queried with temperature 0.0 for deterministic scoring. A representative filtering system prompt is:

Statistic	ISAC	JCC	SAGIN
Raw documents evaluated	~3,500	~8,000	~12,000
Documents retained after filtering	49	132	44
Indexed chunks	1,615	10,350	10,451
Generated QA pairs	678	4,251	3,988
Final train size	610	3,825	3,589
Final validation size	34	213	199
Final test size	34	213	200

Table 3: Filtering and generation statistics for a representative subset of telecommunications domains.

Component	Implementation / Model
Primary generator	meta-llama/Llama-3.3-70B-Instruct
Embedding model	intfloat/e5-large-v2
Refinement filter	GPT-5.2
Vector indexing	FAISS flat index; FAISS IVF-PQ

Table 4: Summary of model usage in the data generation pipeline.

You are a wireless communications and sensing domain expert curating a tutorial dataset. Include explanatory/instructional content about ISAC concepts, such as waveform design, beamforming, and sensing metrics, and exclude marketing or pure radar papers.

Knowledge indexing and chunking. We use the intfloat/e5-large-v2 tokenizer, a BERT-based WordPiece tokenizer, for chunking. Documents are partitioned into chunks of 128 tokens with a 16-token overlap. For training data generation, we use a FAISS flat index with L2 or inner-product similarity for maximum retrieval precision. For large-scale retrieval, we use an IVF-PQ index with $nlist = 4096$ and $m = 64$.

Grounded QA pair generation. QA pairs are generated using a large instruction-tuned language model. The generation uses temperature 0.2, top- $p = 0.9$, and a maximum output length of 512 tokens. One QA pair is generated per 128-token chunk. The generation prompt instructs the model to produce high-quality, self-contained training examples that are answerable solely from the provided text chunk. A representative user prompt is:

From the TEXT_CHUNK below, generate high-quality, self-contained training examples. Each question must be answerable SOLELY using the context. Do NOT reference “this paper” or “above/below”. Include technical reasoning steps in the answer.

Automated retriever alignment. To create gold training pairs for retrieval models, generated questions are mapped back to the chunk index using a weighted alignment score:

$$S = 3.0 \cdot \mathbf{1}_{\text{ID_Match}} + 2.0 \cdot \mathbf{1}_{\text{Contains}} + 1.0 \cdot J_{\text{Tokens}} + 2.0 \cdot J_{\text{Trigrams}}, \quad (3)$$

where J_{Tokens} and J_{Trigrams} denote token-level and trigram-level Jaccard similarity, respectively, between the question’s generation context and the candidate chunk. Chunks are retained as positives if $S \geq 0.05$. For each question, the top $k = 3$ positive chunks are retained.

Negative retrieval examples. For contrastive retriever training, such as training with a Contriever or DPR objective, in-batch hard negatives are constructed dynamically. For each question, the top $k = 20$ chunks are retrieved from the index using the current query encoder, and the highest-scoring chunks that are not in the positive set are selected as negatives. Chunks originating from the same source document as the positive chunk are optionally excluded to increase negative difficulty.

Table 4 summarizes the main models and infrastructure used in the pipeline.

A.2 Baseline Retriever Training Implementations

We implement several baseline retriever training methods. The baseline framework supports retrieval-augmented generation objectives, REPLUG-style supervision, supervised contrastive learning, fixed or scheduled mixed-objective training, and joint retriever-generator adaptation. All baselines use a dense bi-encoder retriever initialized from intfloat/e5-large-v2. Retrieval is performed over the chunk index constructed from the data generation pipeline described in Appendix A.1.

RAG negative log-likelihood objective. The first baseline trains the retriever using a retrieval-augmented generation negative log-likelihood objective. Given a question x , answer y , and retrieved document set D , the loss is

$$\mathcal{L}_{\text{RAG}} = -\log \sum_{d \in D} p_{\eta}(d | x) p_{\theta_0}(y | x, d), \quad (4)$$

where p_{η} denotes the retriever and p_{θ_0} denotes the generator. The generator is kept fixed and provides likelihood scores for the answer conditioned on each retrieved chunk. The retriever is optimized to assign higher probability to chunks that better support generation of the target answer.

Baseline	Description
RAG-NLL	Optimizes retriever using answer likelihood under retrieved chunks
REPLUG-KL	Matches retriever distribution to generator-induced teacher distribution
Contrastive	Trains on aligned positive chunks and non-positive retrieved candidates
Mixed objective	Combines RAG and contrastive losses with fixed or scheduled weights
Joint adaptation	Supports retriever-only, generator-only, and joint retriever-generator updates

Table 5: Summary of the baseline retriever training methods.

REPLUG-style KL objective. The second baseline follows a REPLUG-style objective, where the frozen generator provides a teacher distribution over retrieved chunks. Candidate chunks are scored according to the generator likelihood of the target answer, and the resulting scores are converted into a distribution over passages. The retriever is then trained to match this generator-induced distribution:

$$\mathcal{L}_{\text{REPLUG}} = \text{KL}(Q_{\theta_0}(d \mid x, y) \parallel P_{\eta}(d \mid x)), \quad (5)$$

where Q_{θ_0} is the teacher distribution induced by the generator and P_{η} is the retriever distribution. This objective encourages the retriever to prioritize chunks that the generator finds most useful for answering the question.

Supervised contrastive objective. We also include a supervised contrastive baseline based on an InfoNCE-style loss. Positive chunks are obtained from the retriever-target alignment procedure described in Appendix A.1. Negative chunks are selected from retrieved candidates that are not labeled as positives, together with in-batch negatives. This objective directly trains the retriever to assign higher similarity to aligned positive chunks than to non-positive candidates.

Mixed-objective baselines. The implementation supports mixed-objective baselines that combine the RAG and contrastive losses:

$$\mathcal{L}_{\text{mix}} = \lambda_{\text{RAG}} \mathcal{L}_{\text{RAG}} + \lambda_{\text{cont}} \mathcal{L}_{\text{cont}}. \quad (6)$$

We consider non-adaptive baseline weighting strategies, including fixed weighting and warmup-based scheduling. Fixed weighting keeps λ_{RAG} and λ_{cont} constant throughout training, while warmup scheduling gradually increases the contribution of one objective after the other has stabilized. These baselines allow us to compare single-objective retriever training against simple multi-objective combinations without using the proposed adaptive-temperature regularized formulation.

Joint retriever-generator adaptation baseline. In addition to retriever-only baselines, the framework includes a joint retriever-generator adaptation baseline. This setting supports updating only the query encoder, only the generator, or both components. Generator adaptation is performed using parameter-efficient fine-tuning. This baseline evaluates whether adapting the generator together with the retriever improves retrieval-augmented likelihood relative to retriever-only optimization.

A.3 Models

This section summarizes the language models and embedding models used throughout the experiments. Our setup follows a retriever-centric adaptation framework, where the document encoder is kept fixed, the query encoder is adapted, and the generator is used only for downstream answer generation and evaluation.

Generator models. For the main set of experiments, we use **Llama-3-8B** as the primary generator model. This model is adopted in most experiments because it provides a strong and stable generation backbone for domain-specific RAG evaluation, while remaining practical for repeated training and inference. Using a single strong generator across the main experiments also makes it easier to isolate the contribution of retriever adaptation, since improvements can be attributed more directly to the retriever rather than to changes in generator capacity.

In addition to the main generator, we also evaluate a set of smaller and alternative generator backbones to study how the proposed method behaves across different model scales and model families. Specifically, we use **Llama-3.2-1B**, **Llama-3.2-3B**, and **Qwen3-8B**. The two Llama-3.2 models allow us to examine how the benefit of retriever optimization changes as the generator capacity increases from a small to a medium-scale setting within the same model family. The Qwen3-8B model is included to study whether the behavior of the proposed method is consistent across different model

families at roughly comparable scale. This comparison is important because it helps distinguish gains that are due to retriever optimization from gains that may arise purely from generator architecture or pretraining differences.

Role of model scale in evaluation. The multi-model comparison is intended to assess whether the proposed retriever optimization method remains useful when paired with generators of different strengths. In particular, smaller generators are expected to rely more heavily on the quality of retrieved evidence, since they have less parametric capacity to compensate for missing or imperfect context. Larger generators, by contrast, may already perform strongly under a standard RAG setup, which can reduce the visible margin of improvement attributable to retriever adaptation. Evaluating across Llama-3.2-1B, Llama-3.2-3B, and Qwen3-8B therefore provides a controlled way to study how the effectiveness of the proposed method depends on model capacity and model family.

Embedding model. For dense retrieval, we use `intfloat/e5-large-v2` as the embedding model. This model serves as the common backbone for both document and query representations. In particular, the document chunks are embedded using the base E5 model and indexed once, while the query encoder is initialized from the same base model and then adapted during training. This design ensures that query and document representations begin in the same embedding space and that the adapted retriever remains grounded in the pretrained geometry of the base encoder.

Using the same embedding backbone for both document and query representations is important for two reasons. First, it provides a consistent retrieval space in which similarity comparisons are well-defined. Second, it enables the query-distillation regularizer used in ARMOR, since the adapted query encoder can be directly compared against the frozen base query encoder that shares the same pretrained initialization. In this way, `intfloat/e5-large-v2` acts not only as the initial dense retriever backbone, but also as the reference space that anchors controlled domain adaptation.

Document and query encoding setup. Throughout all experiments, document embeddings remain fixed and are always computed with the base `intfloat/e5-large-v2` encoder. Only the query encoder is updated during retriever training. This query-only adaptation strategy avoids repeated re-indexing of the document corpus and allows us to evaluate whether domain-specific improvements can be obtained by specializing the query side alone. It also makes the regularization design especially important, since the adapted query encoder must remain compatible with the frozen base document space used at inference time.

A.4 Evaluation Harness and Methodology

To evaluate the domain-adapted retrievers and downstream QA models in a consistent and reproducible way, we build an automated evaluation harness over two telecommunications benchmarks: **Tele-Eval** and **TeleQnA**. The harness constructs domain-specific evaluation splits and reports metrics that separately capture retrieval quality and end-task answer quality.

Source benchmarks and domain-specific splits. Our evaluation pipeline uses two complementary benchmark sources. **Tele-Eval** is an open-ended question-answering benchmark for telecommunications. To construct domain-specific test splits, we align benchmark questions with our curated telecom corpus using source-document identifiers. In particular, questions are assigned to a target domain by matching their source documents to the domain-specific document subsets used in our data curation pipeline. This exact document-level mapping ensures that the resulting evaluation questions correspond faithfully to the intended domains, such as ISAC, JCC, and SAGIN.

TeleQnA is a multiple-choice question-answering benchmark designed for telecommunications. Since it is not pre-segmented by our target domains, we construct domain-specific splits using a two-stage filtering procedure. First, candidate questions are matched against domain profiles containing inclusion and exclusion criteria, strong domain-specific terminology, and required telecommunications anchors. Second, a strict language-model-based relevance judge evaluates the domain compatibility of each question-answer pair and retains only high-confidence in-domain examples. This procedure yields domain-specific TeleQnA subsets that are suitable for evaluating transfer beyond the training corpus.

Evaluation metrics. The evaluation harness supports several inference settings, including closed-book generation and retrieval-augmented generation, and reports metrics tailored to each benchmark.

For **Tele-Eval**, we evaluate both answer quality and retrieval fidelity. We report an **Average Score** using an LLM-as-a-Judge protocol, where generated answers are compared against reference answers and assigned a continuous score in $[0, 1]$. Higher values indicate better agreement with the reference in terms of technical correctness, completeness, and usefulness. The final Tele-Eval score is the mean of these per-example scores over the evaluation split. To isolate retrieval performance in RAG configurations, we also report **Recall@K** for $K \in \{1, 3, 5\}$, which measures the fraction of queries for which a relevant source document is retrieved within the top- K results.

For **TeleQnA**, we report **Accuracy**, defined as exact-match multiple-choice accuracy. This metric measures the fraction of questions for which the model selects the correct gold option from the provided answer choices.

Evaluation scope. The two benchmarks serve complementary roles. Tele-Eval is partially aligned with the same underlying corpus used for retriever adaptation and therefore provides a targeted measure of in-domain retrieval and answer generation. TeleQnA, by contrast, is not constructed from the same training corpus and thus provides a stronger test of downstream generalization. Taken together, these benchmarks allow us to assess whether a method improves only retrieval on the training-aligned corpus or also transfers to a broader telecom QA setting.

Summary. Overall, the evaluation harness is designed to disentangle retrieval quality from downstream QA quality. Recall@K measures whether the retriever surfaces relevant evidence, the Tele-Eval Average Score measures the quality of open-ended generated answers, and TeleQnA accuracy measures end-task multiple-choice performance. This structure allows us to evaluate proposed retriever adaptation methods along both retrieval and generation axes using domain-specific telecommunications benchmarks.

B Generalization theory for SFT and RAG fine-tuning

In this section, we derive generalization bounds for generator and retriever fine-tuning, which provides insight into why retriever adaptation may be a better candidate for low-data domains.

B.1 General setup

We compare two families of adaptation, *SFT* and *RAG*, on the same population distribution $\mathcal{P}$ over the data point $z = (x, y)$, where x is the question, and y is the target answer. Given N i.i.d. training triples $\{z_i\}_{i=1}^N$ drawn from the population distribution $\mathcal{P}$, let $\theta \in \mathcal{E}_S \subset \mathbb{R}^{d_S}$ be the trainable model parameter for SFT, $\eta \in \mathcal{E}_R \subset \mathbb{R}^{d_R}$ be the trainable RAG parameters, and $\ell_S(z; \theta)$ and $\ell_R(z; \eta)$ be the per-sample SFT and RAG loss, respectively.

B.1.1 Population and empirical losses

The population SFT loss is defined as the expected test loss under the true population distribution $\mathcal{P}$, while the empirical SFT loss is defined as the empirical training loss on finite sample $\{z_i\}_{i=1}^N$, i.e.

$$\text{population loss: } L_S(\theta) = \mathbb{E}_{z \sim \mathcal{P}} [\ell_S(z; \theta)], \quad \text{empirical loss: } \hat{L}_{S,N}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell_S(z_i; \theta).$$

The corresponding population and empirical risk SFT minimizer are defined as

$$\text{population solution: } \theta^* \in \arg \min_{\theta \in \mathcal{E}_S} L_S(\theta), \quad \text{empirical solution: } \hat{\theta} \in \arg \min_{\theta \in \mathcal{E}_S} \hat{L}_{S,N}(\theta).$$

Similarly, we can define the population and empirical RAG losses and their corresponding optimal solutions as follows.

$$\text{population loss: } L_R(\eta) = \mathbb{E}_{z \sim \mathcal{P}} [\ell_R(z; \eta)], \quad \text{empirical loss: } \hat{L}_{R,N}(\eta) = \frac{1}{N} \sum_{i=1}^N \ell_R(z_i; \eta).$$

$$\text{population solution: } \eta^* \in \arg \min_{\eta \in \mathcal{E}_R} L_R(\eta), \quad \text{empirical solution: } \hat{\eta} \in \arg \min_{\eta \in \mathcal{E}_R} \hat{L}_{R,N}(\eta).$$

The goal of generalization theory is to bound the gap between the population loss and the empirical loss, where a smaller bound indicates better generalization ability of the empirical optimal solution.

B.1.2 Function class and Rademacher complexity

To establish the generalization theory, we rewrite the objectives for SFT and RAG in terms of their backbone prediction model class.

Let $\mathcal{Y}$ denote the answer label space and denote the probability space for RAG answer as

$$q_\eta(\cdot | x) = \sum_{d \in \mathcal{N}_k(x)} p_\eta(d | x) p_{\theta_0}(\cdot | x, d)$$

where $\mathcal{N}_k(x)$ is the top- k document set ranked by the retrieval score $s_\eta(x, d) = f_\eta(x)^\top g(d)$, and

$$p_\eta(d | x) = \frac{\exp(s_\eta(x, d))}{\sum_{d' \in \mathcal{N}_k(x)} \exp(s_\eta(x, d'))}.$$

Therefore, we can write the backbone function class for SFT and RAG as

$$\mathcal{F}_\theta = \{x \mapsto p_\theta(\cdot | x) : \theta \in \mathcal{E}_S\}, \quad \mathcal{F}_\eta = \{x \mapsto q_\eta(\cdot | x) : \eta \in \mathcal{E}_R\}. \quad (7)$$

Letting $\ell_{\text{NLL}}(q, x, y) = -\log q(y | x)$ be the negative log-likelihood loss, the SFT and RAG loss can be rewritten as

$$\ell_S(z; \theta) = \ell_{\text{NLL}}(p_\theta, x, y), \text{ and } \ell_R(z; \eta) = \ell_{\text{NLL}}(q_\eta, x, y).$$

Rademacher complexity. Given the training samples $\mathcal{Z}_N = \{z_i\}_{i=1}^N$, the empirical Rademacher complexity of a function class $\mathcal{F}$ for these samples is defined as follows

$$\mathcal{R}_{\mathcal{Z}_N}(\mathcal{F}) = \frac{1}{N} \mathbb{E}_{\epsilon \sim \text{unif}(\{1, -1\})} \left[\sup_{f \in \mathcal{F}} \sum_{n=1}^N \epsilon_n f(x_n) \right]$$

With Rademacher complexity, we have the following generalization bound theory.

Theorem 2. Assume that the loss function $\ell_{\text{NLL}}(q, x, y)$ is bounded in $[0, c]$ and is ρ -Lipschitz w.r.t q in the feasible domain, then with probability at least $1 - \delta$ over the samples $\mathcal{Z}_N = \{z_n\}_{n=1}^N$, we have

$$\sup_{\theta \in \mathcal{E}_S} \left\{ L_S(\theta) - \widehat{L}_{S,N}(\theta) \right\} \leq 2\rho \mathcal{R}_{\mathcal{Z}_N}(\mathcal{F}_\theta) + 3c \sqrt{\frac{\log(2/\delta)}{2N}} \quad (8)$$

$$\sup_{\eta \in \mathcal{E}_R} \left\{ L_R(\eta) - \widehat{L}_{R,N}(\eta) \right\} \leq 2\rho \mathcal{R}_{\mathcal{Z}_N}(\mathcal{F}_\eta) + 3c \sqrt{\frac{\log(2/\delta)}{2N}} \quad (9)$$

Proof. The proof follows directly from the generalization bounds for machine learning models in terms of Rademacher complexity, e.g. [18, Theorem 3.1], or [1, Theorem B.1]. $\square$

According to Theorem 2, the second term in the generalization bound for SFT and RAG are the same, so we will compare the Rademacher complexity $\mathcal{R}_{\mathcal{Z}}(\mathcal{F}_\theta)$ for SFT and $\mathcal{R}_{\mathcal{Z}}(\mathcal{F}_\eta)$ for RAG in the subsequent sections given the particular model architecture.

B.2 Generalization bound for generator fine-tuning

In this section, we specify the generator model as Transformer-based model. For simplicity, we analyze the one-layer Transformer model following [19], but extension to multi-layer Transformer is also possible [23]. We denote the input for a single question-answer pair (x, y) after tokenization as $X \in \mathbb{R}^{d_S \times d}$, and write the SFT parameter $\theta \in \mathcal{E}_S \subset \mathbb{R}^{d_S}$ as $\theta \triangleq \{W_Q, W_K, W_V, W_O\}$, where $W_Q, W_K, W_V \in \mathbb{R}^{d_m \times d}$, $W_O \in \mathbb{R}^{d_m}$ are the query, key, value and output weight parameters in the Transformer. Then the output of Transformer can be expressed as

$$p_\theta(\cdot | x) = W_O^\top \left(\frac{1}{d_S} \sum_{i=1}^{d_S} \sigma_r \left(W_V X^\top \sigma_s \left(\frac{X W_K^\top W_Q (X^{(i,:)})^\top}{\sqrt{d_m}} \right) \right) \right) \in \mathbb{R}$$

where $X^{(i,:)}$ denotes i -th row of tokenization matrix X for question-answer data pair (x, y) , σ_s denotes the row-wise softmax and σ_r denotes the ReLU activation function.

Following the lazy training regime in [19, 23], we focus on the settings where the Transformer parameters remain close to their initialization throughout training, i.e. the parameters are bounded.

Theorem 3 ([19, Lemma 1]). Suppose $\|W_V^t\|_F \leq R_V$, $\|W_K^t\|_F \leq R_K$, $\|W_Q^t\|_F \leq R_Q$, $\|W_O^t\|_F \leq R_O$ holds for all iteration t . Also assume that the tokenized inputs for (x_n, y_n) have full rank and $\|X_n\|_F \leq \sqrt{d_S} R_X$ for all $n \in [N]$ with positive constant R_X . The empirical Rademacher complexity of the class of Transformer model $\mathcal{F}_\theta = \{x \mapsto p_\theta(\cdot | x) : \|\theta\| \leq R\}$ can be upper bounded by

$$\mathcal{R}_{\mathcal{Z}_N}(\mathcal{F}_\theta) \leq \tilde{\mathcal{O}} \left(\sqrt{\frac{P_S}{N^3}} \left(1 + \log \left(R_O R_V (\sqrt{d_S} R_X) \sqrt{\frac{N}{P_S}} \right) \right) \right) \triangleq C(N, R, R_X, d_S, d_m) \quad (10)$$

where $\tilde{\mathcal{O}}$ hides logarithmic dependencies except $R, N, R = \sqrt{R_V^2 + R_K^2 + R_Q^2 + R_O^2}$, and

$$P_S = (\sqrt{d_S} R_X)^2 \left(\left(\sqrt{d_m} R_V \right)^{\frac{2}{3}} + \left(\sqrt{d_m} R_K R_Q R_V \right)^{\frac{2}{3}} \right)^3 \log(N d_S).$$

Theorem 3 shows that the Rademacher complexity of the Transformer class grows with the parameter dimensions d_m and d_S . For low-data regime where N is small, this dimension-dependent term can dominate Theorem 2, which worsens the generalization bound for SFT.

B.3 Generalization bound for retriever fine-tuning

In this section, we also specify the retrieval model as Transformer model, which is aligned with the model we used in experiments. Similar to the SFT setting, we denote the input for a single question x after tokenization as $X_R \in \mathbb{R}^{d_R \times d}$, and write the retrieval parameter $\eta \in \mathcal{E}_R \subset \mathbb{R}^{d_R}$ as $\eta \triangleq \{U_Q, U_K, U_V, U_O\}$, where $U_Q, U_K, U_V \in \mathbb{R}^{d_{m'} \times d}$, $U_O \in \mathbb{R}^{d_{m'} \times d_E}$ are the query, key, value and output weight parameters in the Transformer. The output of Transformer can be expressed as

$$f_\eta(x) = U_O^\top \left(\frac{1}{d_R} \sum_{i=1}^{d_R} \sigma_r \left(U_V X_R^\top \sigma_s \left(\frac{X_R U_K^\top U_Q (X_R^{(i,\cdot)})^\top}{\sqrt{d_{m'}}} \right) \right) \right) \in \mathbb{R}^{d_E}$$

where $X_R^{(i,\cdot)}$ denotes i -th row of tokenization matrix X_R for question x , σ_s denotes the row-wise softmax and σ_r denotes the ReLU activation function.

Using the learned query-encoder $f_\eta(x)$, the output of RAG is

$$q_\eta(\cdot | x) = \sum_{d \in \mathcal{N}_k(x)} p_\eta(d | x) p_{\theta_0}(\cdot | x, d)$$

where $\mathcal{N}_k(x)$ is the top- k document set ranked by the retrieval score $s_\eta(x, d) = f_\eta(x)^\top g(d)$, and

$$p_\eta(d | x) = \frac{\exp(s_\eta(x, d))}{\sum_{d' \in \mathcal{N}_k(x)} \exp(s_\eta(x, d'))}.$$

Theorem 4. Suppose $\|U_V^t\|_F \leq R_V$, $\|U_K^t\|_F \leq R_K$, $\|U_Q^t\|_F \leq R_Q$, $\|U_O^t\|_F \leq R_O$ holds for all iteration t . Also assume that the tokenized inputs for x_n have full rank and $\|X_{R,n}\|_F \leq \sqrt{d_R} R_X$ for all $n \in [N]$ with positive constant R_X . Additionally, assume $\|g(d)\| \leq B_g$ and q_η is ρ^{soft} -Lipschitz over $s_\eta(x, d)$ on the feasible domain. The empirical Rademacher complexity of the class of retrieval-augmented predictors $\mathcal{F}_\eta = \{x \mapsto q_\eta(\cdot | x) : \|\eta\| \leq R\}$ can be upper bounded by

$$\mathcal{R}_{\mathcal{Z}_N}(\mathcal{F}_\eta) \leq \tilde{\mathcal{O}} \left(\rho^{\text{soft}} B_g \sqrt{d_E} C(N, R, R_X, d_R, d_{m'}) \right) \quad (11)$$

where $\tilde{\mathcal{O}}$ hides logarithmic dependencies except R, N , and $C(N, R, R_X, d_R, d_{m'})$ is defined in (10).

Proof. First, for each coordinate $j \in [d_E]$, let us denote the j -th coordinate of the query embedding as $f_{\eta,j}(x)$ and define the corresponding query-embedding class as

$$\mathcal{G}_{\eta,j} = \{x \mapsto f_{\eta,j}(x) : \eta \in \mathcal{E}_R\}$$

Then applying Theorem 3, we know that for any $j \in [d_E]$,

$$\mathcal{R}_{\mathcal{Z}_N}(\mathcal{G}_{\eta,j}) \leq \tilde{\mathcal{O}}(C(N, R, R_X, d_R, d_{m'})) \quad (12)$$

Next, define the scalar retrieval score as

$$\mathcal{S}_{\eta,i} = \{x \mapsto f_\eta(x)^\top g(d_i) : \eta \in \mathcal{E}_R\}$$

where $i \in [k]$ and $d_i \in \mathcal{D}_k(x)$. Then, since $f_\eta(x)^\top g(d_i) = \sum_{j=1}^{d_E} f_{\eta,j}(x)g_j(d_i)$, and by the Cauchy-Schwarz inequality, we have

$$|f_\eta(x)^\top g(d_i)| \leq B_g \left(\sum_{j=1}^{d_E} f_{\eta,j}(x)^2 \right)^{1/2} \quad (13)$$

Therefore, according to the definition of Rademacher complexity, we have

$$\mathcal{R}_{\mathcal{Z}_N}(\mathcal{S}_{\eta,i}) \leq B_g \left(\sum_{j=1}^{d_E} \mathcal{R}_{\mathcal{Z}_N}(\mathcal{G}_{\eta,j})^2 \right)^{1/2} \leq \tilde{\mathcal{O}}(B_g \sqrt{d_E} C(N, R, R_X, d_R, d_{m'})) \quad (14)$$

Finally, applying Talagrand's lemma for composite mappings [18, Lemma 4.2] to the softmax-weighted generator mixture gives

$$\mathcal{R}_{\mathcal{Z}_N}(\mathcal{F}_\eta) \leq \tilde{\mathcal{O}}(\rho^{\text{soft}} B_g \sqrt{d_E} C(N, R, R_X, d_R, d_{m'})) \quad (15)$$

□

Remark 2. Because of top- k retrieval, the original $q_\eta(\cdot | x)$ might not be continuous over η , so it might not satisfy Lipschitz continuity over $s_\eta(x, d)$. Following [3], we approximate $q_\eta(\cdot | x)$ by a soft surrogate with a fixed document pool $\mathcal{D}_k(x)$ as

$$q_\eta(\cdot | x) = \sum_{d \in \mathcal{D}_k(x)} p_\eta(d | x) p_{\theta_0}(\cdot | x, d)$$

where

$$p_\eta(d | x) = \frac{\exp(s_\eta(x, d))}{\sum_{d' \in \mathcal{D}_k(x)} \exp(s_\eta(x, d'))}.$$

With fixed k document pools for given x , q_η can be Lipschitz continuous over s_η [3].

B.4 Comparisons between generator and retriever fine-tuning

Combining Theorem 2–4, we proved Theorem 1 and we know that when the number of samples N is small ($N \ll d_R, d_S$), and the number of parameters of SFT model fine-tuning is far more than the retrieval tuning, i.e. $d_R \ll d_S$, the empirical Rademacher complexity dominates the generalization bounds for both SFT and RAG. However, RAG has smaller generalization bound than the SFT thanks to the small dimension induced empirical Rademacher complexity.

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